CLASS 9 SCIENCE · QUICK NOTES

CHAPTER 6

How Forces Affect Motion

Force · Balanced & unbalanced forces · Friction · Newton's three laws · System of objects

THE CHAPTER IN 30 SECONDS

A force (push or pull) changes motion. What matters is the **net force**. **Zero net force** → velocity does not change (1st law). **Non-zero net force** → acceleration **a = F ÷ m** (2nd law). Forces always come as **equal and opposite pairs acting on two different objects** (3rd law).

1 The concept of force

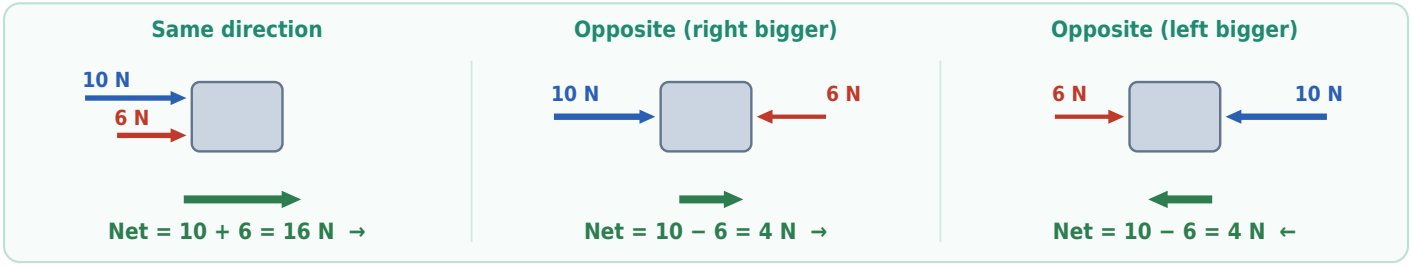
NCERT 6.1

- A force is a **push or pull**. It can start or stop motion, change speed, change direction, or change shape (squeezing a lemon).
- Force is a **vector**: it needs magnitude **and** direction. SI unit: **newton** (written with a small 'n'), symbol **N**.
- Measured with a **spring balance**. The **weight** of an object is the gravitational force the Earth exerts on it.
- Change the magnitude or the direction of a force and its effect changes. A 100 g mass resting on your palm presses with about **1 N**.

2 Balanced and unbalanced forces

NCERT 6.2

	Balanced forces	Unbalanced forces
Forces	Equal in size, opposite in direction	Not equal (or not opposite)
Net force	Zero	Non-zero
Effect	No change in motion (rest stays rest; moving keeps its velocity)	Velocity changes: speeds up, slows down or turns
Example	Tug of war with equal pulls: rope does not move	One team pulls harder: rope moves toward it



★ MEMORY HOOK

Same direction → add. Opposite directions → subtract (net force points along the bigger force).

REMEMBER

Many forces may act on an object, but its motion depends **only on the net force**. A floating ball: weight (down) = buoyant force (up). A book on a table: weight (down) = normal force (up).

3 Friction: always present

NCERT 6.3

- Friction acts **between surfaces in contact**, in the direction **opposite** to the motion (or to the applied force).
- A box starts to move only when the applied force becomes **larger than friction**. To keep it moving at constant velocity, the push must **equal** friction.
- When you stop pushing, a moving object slows down and stops **because of friction**, not because motion "runs out".
- **Activity result (stack of coins):** wood < laminate < marble in distance travelled. Smoother surface → **smaller friction** (smaller spring-balance reading) → coins slow down more gradually and travel farther.
- **Galileo's thought experiment:** with zero friction on an endless smooth floor, the object would **keep moving forever** at constant velocity.

- **Friction helps:** walking, running, grooves on shoe soles, treads on tyres. Wet floors and ice are slippery because friction is small. Air also exerts friction (usually small).

4 Newton's first law (law of inertia)

NCERT 6.4

An object at rest stays at rest, and an object in motion keeps moving with constant velocity, unless a net force acts on it.

- Net force = 0 ⇒ **acceleration = 0**. **Inertia** = the tendency of an object to resist a change in its state of rest or uniform motion (Newton's word).
- "Constant velocity" means no change in speed **or** direction (a straight line at the same speed).
- A force is needed only to **start, stop or change** the velocity, not to keep an object moving at constant velocity.

Object with zero net force	x-t graph	v-t graph
At rest	horizontal line	on the time axis (v = 0)
Constant velocity	straight inclined line	horizontal line

✎ Example 6.2
You push a moving box with a force equal to the friction on it. Net force = 0, so the box **keeps moving at constant velocity**.

5 Newton's second law

NCERT 6.5

a = F ÷ m
F = ma
1 N = 1 kg m s^{-2} (force that gives 1 kg an acceleration of 1 m s^{-2})

weight F = mg
g = 9.8 m s^{-2} (use 10 for quick estimates) · independent of mass

- A net force makes the object accelerate **in the direction of the net force**. **a ∝ F** (double the force → double the acceleration). **a ∝ 1/m** (double the mass → half the acceleration).
- **Cart and pulley experiment:** from rest $s = \frac{1}{2}aT^2$, so $a_1 \div a_2 = T_2^2 \div T_1^2$. Results differ slightly from exact doubling because of friction and measurement errors.
- Complete form (higher grades): F = rate of change of momentum, where momentum p = mv.

Everyday example	What happens (F = ma, same change in velocity)
Fielder pulls hands back while catching	Longer stopping time → smaller acceleration → **smaller force**, less injury
Airbag and seat belt	Passenger stops over a longer time → smaller force on the body
High-jump landing mat / sand bed; bubble wrap or hay for glass	Lengthens the stopping time → smaller force
Cracking a coconut on hard ground	Stops in a very short time → **very large force** → shell breaks

★ MEMORY HOOK
To make a force smaller, make the stopping time longer. Same change in velocity, more time → smaller acceleration → smaller force (F = ma).

✎ Barbell (6.4)
10 + 10 + 10 = 30 kg. F = mg = 30 × 9.8 = **294 N** upward to hold it steady.

✎ Block (6.5)
25 kg, friction 50 N.
Push 50 N → balanced, stays still.
Push 55 N → net 5 N, a = 0.2 m s^{-2}, s in 2 s = ½ × 0.2 × 4 = **0.4 m**.

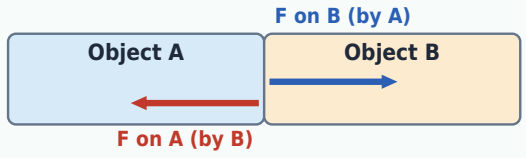
✎ Sports car (6.6)
1500 kg. 0–5 s: a = 2 → **3000 N** east. 5–10 s: constant v → **0 N**. 10–15 s: a = −2 → **−3000 N** (west).

6 Newton's third law

NCERT 6.6

Whenever one object exerts a force on a second object, the second object exerts an equal and opposite force on the first.

Third law: a pair acts on TWO different objects



equal size · opposite direction · never cancel

KEY POINTS

- Forces come in **pairs**, at the same moment.
- They act on **different objects**, so they **never cancel**. (Equal and opposite forces on the *same* object do balance.)
- True for contact **and** non-contact forces (magnets, charged balloons, gravity).

Situation	Force 1	Force 2 (its pair)
Walking / running	Foot pushes the ground backward	Ground's friction pushes you forward
Rowing a canoe	Paddle pushes water backward	Water pushes paddle and canoe forward
Climbing a tree	Legs push the trunk down	Friction pushes the person up
Rocket / balloon	Engine pushes exhaust gas down (air out)	Gas pushes rocket up (balloon forward); Chandrayaan-3 lander slowed this way
Chair on wheels and table	You push the table forward	Table pushes you (and the chair) backward
Earth and a falling fruit	Earth pulls the fruit down	Fruit pulls Earth up (Earth's acceleration is tiny: huge mass)

✎ **Gun and bullet (6.8)**

Force on each = 2 N. Gun 5 kg: a = 2 ÷ 5 = **0.4 m s⁻²**. Bullet 0.1 kg: a = 2 ÷ 0.1 = **20 m s⁻²**.

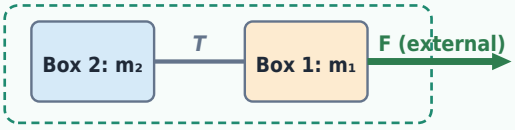
★ **MEMORY HOOK**

Equal forces ≠ equal accelerations. The pair is equal, but a = F ÷ m, so the lighter object accelerates more.

7 Forces on a system of objects

NCERT 6.7

System of two boxes (frictionless surface)



a = F ÷ (m₁ + m₂)

METHOD

- Treat connected objects as **one system**.
- **Internal forces** (the tension T) are ignored; only **external forces** (F, weight, normal force) matter.
- Weight is balanced by the normal force, so F decides the motion.
- Result: the system accelerates like one object of mass $m_1 + m_2$.

✎ **Quick check (derived from the same idea)**

m_1 = 3 kg, m_2 = 2 kg, F = 20 N on Box 1. System: a = 20 ÷ (3 + 2) = **4 m s⁻²**. Tension pulls only Box 2: T = m_2a = 2 × 4 = **8 N**. Check on Box 1: F − T = 20 − 8 = 12 N = 3 × 4 ✓.

✎ Solved textbook problems

Revise, Reflect, Refine

Problem	Working	Answer
Snake boat: 95 oarsmen row backward, 5 by mistake forward, each 200 N. Net force?	95 × 200 − 5 × 200 = 19000 − 1000	18000 N forward
10 kg object: v–t graph goes from 10 m s^{-1} (0 s) to 30 m s^{-1} (8 s). Force?	a = (30 − 10) ÷ 8 = 2.5 m s^{-2} ; F = ma = 10 × 2.5	F = 25 N
Bullet 50 g at 100 m s^{-1} stops after 0.5 m in a block. Stopping force?	$v^2 = u^2 + 2as \rightarrow 0 = 10^4 + 2a(0.5) \rightarrow a = -10^4$ m s^{-2} ; F = 0.05 × 10^4	F = 500 N
Footballer: 108 km h^{-1} = 30 m s^{-1}, force 800 N, ball 0.4 kg. Contact time?	a = F ÷ m = 800 ÷ 0.4 = 2000 m s^{-2} ; t = v ÷ a = 30 ÷ 2000	t = 0.015 s
2 kg object at 10 m s^{-1} meets friction 7 N plus an extra 3 N against it. Distance to stop?	Total opposing force 10 N → a = −5 m s^{-2} ; s = u^2 ÷ 2|a| = 100 ÷ 10	s = 10 m

★ Quick revision

one-page summary

Idea	Rule / formula	Unit
Net force	same direction: add · opposite: subtract (along the larger)	N
First law	net force = 0 ⇒ a = 0 (rest stays rest, motion stays constant velocity)	
Second law	a = F ÷ m ; F = ma ; weight = mg (g = 9.8 m s^{-2})	N = kg m s^{-2}
Third law	pair: equal, opposite, on two different objects	N
System of objects	a = F ÷ (m_1 + m_2), internal forces ignored	m s^{-2}
Friction	opposes motion; smoother surface → smaller friction	N

10 POINTS TO REMEMBER

1. Force is a **vector**; SI unit newton (N); measured with a spring balance.
2. **Balanced forces** (equal, opposite) have zero net force and do not change motion.
3. Many forces can act, but motion depends only on the **net force**.
4. Friction acts **opposite to motion**; less friction → object travels farther.
5. **1st law:** no net force → rest stays rest, motion stays constant velocity. **Inertia** resists change.
6. **2nd law:** F = ma; acceleration is along the net force, ∝ F and ∝ 1/m.
7. 1 N = the force that gives 1 kg an acceleration of 1 m s^{-2}; weight = mg.
8. Longer stopping time → smaller force (catching a ball, airbags, landing mats).
9. **3rd law:** action and reaction act on **different** objects, so they do not cancel.
10. A connected system accelerates at F ÷ (total mass); internal tension is ignored.

✗ COMMON MISTAKE

- "A moving object needs a continuous force." No: a force is needed only to overcome friction or to change velocity.
- "Action and reaction cancel." No: they act on two different objects.
- "Equal forces give equal accelerations." No: a = F ÷ m depends on mass.
- Using grams instead of kilograms in F = ma; forgetting that friction can also help (walking).

TEST YOURSELF (cover the green column)

Q1. Forces of 10 N (right) and 6 N (left) act on a block. Net force?	**4 N to the right**
Q2. A 5 kg block feels a net force of 20 N. Find its acceleration.	**a = 20 ÷ 5 = 4 m s^{-2}**
Q3. Find the weight of a 3 kg object (take g = 10 m s^{-2}).	**F = mg = 30 N**
Q4. Why does a rocket lift off?	**Engine pushes gas down; gas pushes rocket up (3rd law)**
Q5. An object moves at constant velocity. What is the net force on it?	**Zero (1st law)**